

The *Yijing* / Binary System Episode has haunted scholars repeatedly during the last fifty years. The episode involved the remarkable discovery by Leibniz (and an associate) of the binary geometrical progression in a diagram associated with the *Yijing* that was said to have been created by the mythical third-millennium father of Chinese writing, Fu Xi. A correspondent in Beijing sent Leibniz a copy of the diagram, which depicted yin and yang, apparent symbols of zero and one, in the order of Leibniz' binary number system (Table 1). Leibniz thought he had discovered evidence of a forgotten mathematical science in the Chinese past, and, in spite of his sinological knowledge, he never found evidence to the contrary. Thus, it has been left to contemporary scholars to explain the apparent correspondence.

While the general trend in scholarship has rightly presumed that no forgotten mathematical science existed in ancient China, the conclusion that the *Yijing* / Binary System Episode was a mere coincidence has perhaps not satisfied scholars, in view of the intricacy of the binary geometrical progression and the temporal and spatial isolation of the Chinese Diagram from the European.¹

I will offer a qualification to this conclusion. What Leibniz meant when he attributed "science" to the Diagram² differs from corresponding ideas in a Song dynasty context. However, the principal exponent of the Diagram (Plate I), Shao Yong (1012–1077),³ used it in proto-scientific fashion and grounded this use in metaphysical views similar to those of Leibniz. Shao and Leibniz shared the view that the cosmos was created according to the binary system, which is reflected in all things and discoverable by human beings. Shao's use of the Diagram resembled some of the basic themes of Leibniz' philosophy of science. Nevertheless, Shao stopped short of pushing his Diagram toward a scientific number system, since he believed that rigorously scientific thinking was perniciously dualist and spiritually detrimental. With the picture filled out in this way, we will be able to draw some revised conclusions about the reasons for the haunting coincidence of the *Yijing* / Binary System Episode.

Leibniz and Bouvet's Discovery⁴

Leibniz' interest in China was part of a broader project. He wanted to show the results of all human inquiry, of whatever discipline or culture, to be mutually consistent and to reflect his own metaphysical views. He gathered technological advances from abroad, so that he could advance his goal of an interconnected scientific community. He sought information about non-European languages in an effort to determine the common origin of all languages, and he intended to develop a universal,

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Table 1. The Binary Number System.

A: Binary numbers with denary equivalents		B: Squares	
0	0	0000	0
1	1	0001	1
10	2	0100	4
11	3	10001	9
100	4	10000	16
101	5	11001	25
110	6	100100	36
111	7	110001	49
1000	8	1000000	64
1001	9	Column 1: Period of 01	
1010	10	Column 2: Period of 0	
1011	11	Column 3: Period of 0010	
1100	12	Column 4: Period of 00010100	
1101	13	Etc.	
1110	14	C: Cubes	
1111	15	000	0
10000	16	001	1
10001	17	1000	8
10010	18	11011	27
10011	19	1000000	64
10100	20	1111111	125
10101	21	11011000	216
10110	22	Column 1: Period of 01	
10111	23	Column 2: Period of 0001	
11000	24	Column 3: Period of 00000101	
11001	25	Etc.	
11010	26		
11011	27		
11100	28		
11101	29		
11110	30		
11111	31		
100000	32		

analytical language ("universal characteristic") that would allow fluent and rigorously exact international scientific communication.⁵

Leibniz turned to China in an effort to tap into its thousands of years of cultural development. His sinological studies, extending throughout the second half of his life, probably flowed from his interest in pictographic writing, his project of collecting foreign technological innovations, and his admiration for Confucian moral thought.⁶ Leibniz could have found no more agreeable ally than Joachim Bouvet (1656–1730).⁷ Bouvet, a Jesuit missionary in Beijing, was both trained in mathematics

and science and effusively visionary in his devotion to the Figurist ideal of a universalistic coalescence of the various cultures, religions, and sciences of the world (under the ultimate patronage of Christianity). Bouvet, like many Figurists, believed that Chinese history, since it was exceedingly well recorded, was a promising source for locating predictions of Christ and prescient intimations of Christian mysteries.⁸

It was Bouvet who discovered the correspondence between the *Yijing* and Leibniz' binary system and who first theorized that the Diagram was a remnant of forgotten ancient Chinese science. In order to gain the favor of the Kangxi emperor (r. 1662–1722) for the Jesuit mission, Bouvet wanted to appeal to what he thought were intimations of Christian doctrine in the Chinese classics. Thus, he was happy to find another basis for his claims in the correspondence between the ancient Chinese symbols and the ideas of the modern Christian thinker Leibniz.

In a 1701 letter, Leibniz explained the binary system (shown in Table 1) to Bouvet in some detail. In a response, Bouvet demonstrated the following correspondence to Leibniz. Taking broken lines to be equivalent to zeros and solid lines to be equivalent to ones, the first hexagram at the bottom of the Diagram (䷁, see Plate 1) seems to denote zero as the binary system would. If one overlooks the fact (of which neither Bouvet nor Leibniz was aware) that the Chinese read the hexagram lines from the bottom up, rather than from the top down, the rest of the binary numbers proceed in the Diagram as follows. Reading counterclockwise, ䷁ = 1, ䷂ = 10, ䷃ = 11, ䷄ = 100, ䷅ = 101, and so forth. As Leibniz indicates, the binary system can be seen starting at the bottom of the circle, proceeding counterclockwise to the top and then again clockwise from the bottom to the top.⁹ The double geometric progression can be discerned in the Diagram as well as in Leibniz' binary system. That is, the broken lines (yin) are gradually filled in (replaced with unbroken yang lines) from the outside to the inside. Beginning at the bottom and moving counterclockwise, the first hexagram (䷁) has five innermost broken lines; the next two hexagrams have four innermost broken lines (䷂, ䷃); the next four have three innermost broken lines; the next eight have two; and the next sixteen have one. A similar pattern continues from the bottom clockwise to the top.

Bouvet took Leibniz' work to be a confirmation of his suspicion that the Diagram represented forgotten ancient science. If Western scientific ideas were discovered in their own ancient tradition, the Chinese would open their ear to Western religious ideas.¹⁰ Leibniz agreed that the Diagram was "one of the most ancient monuments of science . . . more than 4000 years old."¹¹ The binary system was, for Leibniz, a representation of creation *ex nihilo* by God, in which each thing is constituted by some permutation of one (i.e., God) and nothing. Thus, Leibniz supported Bouvet's mission to use the binary system and the *Yijing* "to suggest to

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the Chinese the most important truths of philosophy and natural theology in order to facilitate the way to be revealed."¹²

The "introduction of philosophy," which would bring the Chinese "spirits closer to receiving the true religion," involved the same metaphysical agenda as that which Leibniz was putting forth in Europe against the Cartesians.¹³ Leibniz wished to combat materialism in Europe and China by proving that "mechanics implies consideration of force."¹⁴ In order to accomplish this, he intended to "rehabilitate the philosophy of the ancients"¹⁵ of both traditions. For Leibniz equated his concept of the sources of force, the monads, with the Confucian concept of principle (*li*), which he mistook to be an ancient, rather than a relatively recent (Song dynasty), idea in Chinese philosophy.¹⁶

By appealing to the Chinese's original understanding of force in general and to the original force—God's creation of the world *ex nihilo*—Leibniz hoped to bring China into the international scientific community. For Leibniz, *taiji* (the Great Ultimate, the source of the yin and yang) involved the concept of a primordial force from which all other forces ("spirits," *shen*) spring. Thus, the Chinese had some of the concepts necessary for scientific thinking, as well as what Leibniz took to be a better practical knowledge of force than Europe, manifest in its longer history of invention and practical sciences, such as medicine and geography.¹⁷

Leibniz was trying to develop the aforementioned universal characteristic, a "language of thought,"¹⁸ which would be structured grammatically in correspondence to facts in the world. Its words would embody real definitions, combinations of units that reflected the actual constitution of the things to which they referred.¹⁹ Most thinking could therefore be reduced to mechanically following truth-preserving rules for manipulating objective symbols, thereby facilitating international communication and accelerating the progress of science.²⁰ Leibniz planned to pique the Chinese interest in such a community. On the principle that "it will always be found that one truth concurs with the other,"²¹ Leibniz believed that the ancient Diagram and its interpretation by seventeenth-century Chinese²² could be interwoven with his binary system to "give us an open field for devising a new Characteristic which will appear to be a consequence of Fu Xi's [characteristic]."²³

The Diagram: A Scientific Binary Number System?

During the *Yijing* / Binary System Episode, Leibniz made theoretical and historical claims with regard to the Diagram. The theoretical claim was that it was a representation of scientific knowledge of the binary system, which he "rediscovered some thousands of years later."²⁴ The historical claim was that the Diagram had been a scientific instrument of the Chinese thousands of years before, although its significance had

evidently been forgotten in China by the seventeenth century. This claim has been shown to be false, since the Diagram was not so old (perhaps evolving around the first century B.C.).

The evaluation of the theoretical claim—that the Diagram, whenever it was created, was scientific—entails examining what Leibniz meant by a “scientific”²⁵ use of the binary system and comparing that meaning with the Chinese use of the Diagram. I think Leibniz’ theoretical claim has been prematurely dismissed, because much of the proto-scientific thought of the Diagram’s latter-day exponent, Shao Yong, has been overlooked. Shao tried to explain natural change with the Diagram, and he did so in a proto-scientific manner that utilized metaphysical underpinnings similar to Leibniz’ views. Lest one think that certain anti-scientific passages in Shao (which I will discuss) negate the comparison with Leibniz, we note that Shao was considered radically mechanistic and fatalistic by his Neo-Confucian contemporaries.²⁶

Leibniz conceived of science as an organized and quantitative method of observing nature and of deriving and verifying natural laws. While he thought that natural change was in principle fully explainable a priori, he nevertheless argued that experience was the only actual source of explanations open to human beings. Hence, he tended to integrate rationalist and empiricist philosophical approaches to science. As we will see, Shao’s passion for empirical science was not as strong as that of Leibniz. But he shared some of Leibniz’ scientific views, and both thinkers took some of the direction for their thinking from religious and metaphysical views that bear significant similarities.

For Leibniz, since every thing or event has a sufficient reason (a complete explanation of the causes from which its existence follows), contingent propositions are knowable a priori for anyone who has intuitive knowledge of the primitive notions from which things are explained. At one point, Leibniz believed that a genius could in principle deduce truths of science a priori, from the a priori principles he knows, such as the principle of noncontradiction and the nature of God’s perfections (the ultimate reasons for all things).²⁷ But he ultimately came to believe that such an analysis requires an infinite number of steps, which is impossible for finite beings to complete. Leibniz concluded that only God could see the whole infinity of terms at once.²⁸ For only God knows all there is to know about a thing and, thus, why the application of specific natural laws is contained in its complete notion. Since he wanted to create the best possible world, God chose to actualize the one which had the most simply organized and plentiful amount of being.²⁹ Only God knew why the truth of a contingent proposition optimizes the perfection—variety and simplicity—of creation.³⁰ For example, only God could see that the containment in the concept of force of mass multiplied by acceleration optimizes the order and variety of the world in every re-

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spect. A human being would never be able to reduce " $F = ma$ " to an identity. But scientists can infer this equation, since it is the simplest of the possible equations that account for the given finite number of observations.

The only way to knowledge about natural laws that is open to humans is conjecture, which has both a priori and a posteriori aspects. Conjecture a priori involves making hypotheses about a natural law based on the metaphysical view that the true law will be the simplest explanation of the greatest number of objects under study. For on this view, that law can be known a priori to cover the greatest variety of instances. For example, Snell's law (that the sine of the angle of incidence of a light ray is proportional to the sine of the angle of its refraction) was inferred to be correct, since it both fit any given observation of refraction and represented the assumption that light will take the easiest path through two media of unequal resistances. For the proportion was simply given by the ratio of the resistances of the media.³¹

Yet, this inference could not have been made without those observations. Conjecture a priori always occurs along with inductive and analogical reasoning, the verification of which reaches moral, and not deductive, certainty.³² In other words, conjecture a priori is always accompanied by conjecture a posteriori. But even if general truths about nature aren't known as certainly as the truths of arithmetic, the principles which guide conjecture a priori are, nevertheless, mathematically defined.³³ A priori principles of generality, economy, and simplicity, which guide hypotheses, are defined in terms of mathematical concepts of maximum and minimum quantities of order and variety.³⁴

Moreover, empirically gathered primary truths always come organized in quantitative categories. "Since we may perceive nothing accurately except magnitude, figure, motion, and perception itself, it follows that everything is to be explained through these four."³⁵ So, conjecture a priori and conjecture a posteriori, as well as God's certain a priori knowledge of all things, all involve numbers and mathematical truths in a central way. "Number is thus a basic metaphysical figure, as it were, and arithmetic is a kind of statics of the universe by which the powers of things are discovered."³⁶

Leibniz thought that a universal characteristic would make thinking be like calculation by accurately reflecting the numerical structure of nature and organizing the various permutations of primitive kinds (each of which has a corresponding number) according to numerical series. The art of combinations, employing the universal characteristic, "fixes, abridges, and orders"³⁷ the thoughts of the scientist. Mungello points out that while Leibniz' hopes seem far-fetched to us, he was writing during a period in which many intellectuals were seriously searching for such a universal language.³⁸ Moreover, the utility of Leibniz' binary number

system in computer science is, to borrow Mungello's phrase, "very impressive though much more confined in scope than [Leibniz'] claims."³⁹ For Leibniz, this art would guide all scientific inquiry:

[A]fter the isolated experiments are completed, as many experiments as possible are to be tried with only two kinds taken together, as with saltpeter and common salt alone, treating them in various ways by fire, water, air; and by combining various products thus made. . . . After combinations, one should go on to the conternations, etc., or to the ternions, and quaternions of the classes.⁴⁰

So, conjecture a posteriori should rely on the organizational techniques of the art of combinations. And inferences based on experimental data also involve the art of combinations in the form of inferences that follow the geometrical method.⁴¹ From clear, empirically constructed definitions, inferences to general natural laws could be made by analyzing the combined parts of the definitions. Just as a clear definition of a triangle yields by analysis the rule that every triangle's interior angles will sum to 180 degrees, so, too, could one infer natural laws from knowledge of the things involved in a given situation. For instance, by considering the definition of a given complex substance, based on organized observational data, a scientist could deduce whether that substance is flammable.

The number system that most naturally expresses the orderly permutations of two kinds, three kinds ("conternations"), four kinds ("quaternions"), and so forth is the binary system, since it embodies the base-two progression that runs through these permutations. Leibniz demonstrated this periodicity in his early *Dissertation on the Art of Combinations* (1666), particularly in a table of the numerical permutations generated according to the double geometric progression.⁴² I will not describe the mathematical detail involved in the table, but suffice it to say that the table was supposed to make the scientist's job of finding all possible combinations of any number of elements a mechanical-decision procedure, the simplicity of which derives from the periodic series of the binary number system. These periodic series of squares and cubes are shown in Table 1.

Leibniz thought the binary system reflected the principles of simplicity and economy, which, he had shown, were so fundamental to science. It generated all possible permutations with a continuous pattern constructed according to a simple rule. By the art of combinations the scientist would make educated conjecture a priori about what sort of thing will have what sort of behavior. And the scientist would organize the experimentation done under many permutations of conditions.

For Leibniz, then, science involves the rigorously methodical application of quantitative methods to the verification, observation, and deduction of natural laws. His example of experimenting with saltpeter

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shows what this would look like. By experimenting with a finite number of methodically permuted combinations of saltpeter and other substances, a scientist would be better able to draw conclusions about general laws about the behavior of saltpeter than she could by combining saltpeter with random groups of arbitrarily chosen substances. For, by Leibniz' method, the scientist covers the possible permutations in order—combinations of two substances, three substances, and so forth—thus making it possible for her to discern patterns in the behavior of saltpeter. Assuming nature to be simply designed, the scientist need not cover each of the infinitely possible permutations. Rather,

Those experiments are to be tried first in which the analogy to other experiments promises some special effect. It is likely that things which are similar in many known factors will also be similar, or at least approximately so, in other factors not yet examined.⁴³

These organizational methods are centered on numbers, most naturally the binary system, since it reflects the generation of things from one ("God") and zero ("nothing"), guides hypothesis and organizes experimentation according to periodic permutations, and embodies the simplicity and economy of calculation so valuable to science. So, there is a fairly specific sense in which Leibniz attributes ancient science to the Diagram, itself a *de facto* representation of the binary system.

It has been demonstrated, however, that the Diagram did not have a forgotten scientific source in the Third Millennium B.C.⁴⁴ Rather, it has been shown that it had its primary exponent in Shao Yong (eleventh century) and had its origins probably only a few centuries before him. Shao, perhaps the most concerned with numbers and an understanding of natural phenomena of any Neo-Confucian, is not known as an experimental scientist. For Shao, the *Yijing* was a numerologically powerful text. We usually think of numerology as the doctrine that numbers are powerful in the sense of having a supernatural effect upon or control over events. On the other hand, for Leibniz, numbers were scientifically powerful in the sense of representing a large amount of information in a generalized and organized manner.

Instead of paying attention to the mathematical series in the Diagram, Shao seems to have been interested in the correlative metaphysical explanations of the Diagram, analogies between natural bodies and processes and the binary structure of the Diagram. The sixty-four hexagrams in the binary system order are derived from the division of the original Absolute, the "Great Ultimate" (*taiji*), into the two complementary forces of creation, yin and yang—the female and male, the dark and light, the passive and active, and so forth. These two forces (*liangyi*) subdivide until all things are produced by their permutations (Figure 1). "Thus, one divides to make two. Two divides to make four. Four divides

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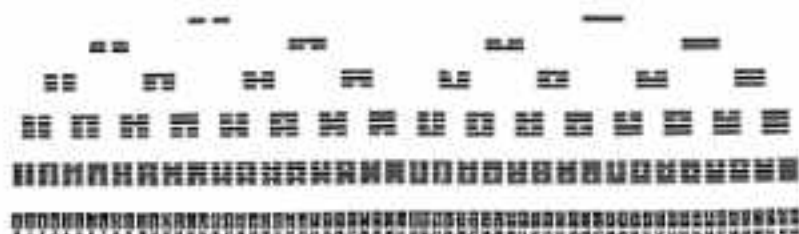


Fig. 1. Progression of the sixty-four hexagrams in the binary system.

to make eight . . . and thirty-two divides to make sixty-four."⁴⁵ While Leibniz had focused on powerful periodic series in developing the binary system, the Chinese seem not to have noticed these but to have generated the Diagram through vertical divisions. Since it is not by chance that the repeated divisions produce the binary system, the *Yijing* / Binary System Episode has been considered a mathematically unsurprising coincidence.

Furthermore, the Chinese counted the innermost line of each hexagram in the circle of the Diagram as the first, whereas the Diagram's correspondence to the binary system implies that the outermost line of each hexagram is the first.⁴⁶ But the geometrical progression begins on the outermost line, only gradually filling out the innermost lines. For all of these reasons, then, scholars have concluded that the resemblance between the binary system and the Diagram was "purely formal."⁴⁷

But there is evidence that this resemblance was not purely a coincidence explainable by purely formal properties of the number two. There is more resemblance between the Diagram's significance in Shao's system and the binary system's significance for Leibniz than has been acknowledged. Shao derived the Diagram using mathematical techniques beyond the mere repeated division of yin and yang. He used the periodic series of the progression to explain the "waxing and waning" of natural phenomena. And he engaged in something similar to conjecture a priori, and even utilized some observation. He partially fulfilled Leibniz' criteria for scientific thinking.

Shao and Leibniz on Metaphysics and Science

There is merit in Leibniz' case that the Diagram reflected his own view of science. Both Leibniz and Shao relied to some extent on correlative thinking,⁴⁸ since both based their theories at least partly on correlations among natural phenomena, numbers, and ideas. That is, Leibniz' way of doing philosophy was not entirely analytical, but to some extent drew upon powerfully intuitive associations—correlations—among metaphysical ideas and natural phenomena. Shao's philosophical aims were not entirely nonscientific; he used his system to attempt a scientific ex-

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planation of natural phenomena (though this attempt was limited, for reasons explained below). And although the scientific power of Shao's system was critically limited by its lack of quantitative and empirical methods, it did represent, through its double geometric progression, something like the conjecture a priori of Leibniz' early-modern view of science. These affinities tend to weaken the contrast between the two thinkers.

- The correlative thinking shared by Leibniz and Shao gives an indication of common interests. Both emphasized creation *ex nihilo* (or in Shao's case, at least creation for the formless material force, *qi*) and considered the binary system to be its symbol. Leibniz took his concept of creation to be a reason for the explanatory power of the binary system, since it

gives an admirable representation of the Creation. It is that following this method, all numbers are written by the co-mingling of the unit and zero, much the same as all creatures coming uniquely from God and nothingness. ... It is not said in vain that all essences are like numbers. ...⁴⁸

Leibniz saw things, essences, and natural laws as structurally analogous to their sufficient reason, their primordial creation. Thus, he believed that thought and language could and should be perfectly correlated with the binary structure of nature. Moreover, the harmonious correlations that numbers exhibit are not explainable, but instead are themselves the rock-bottom basis of all explanations of things. Numerical harmonies form the basic structure of all ideas of things that even God could not have made to be otherwise but was logically compelled to follow in his creation of the cosmos, Leibniz says:

God wills the things which he understands to be best and most harmonious. ... What then is the reason for the divine intellect? The harmony of things. What the reason for the harmony of things? Nothing. For example, no reason can be given for the ratio of 2 to 4 being the same as that of 4 to 8, not even in the divine will. This depends on the essence itself, or the idea of things. For the essences of things are numbers.⁴⁹

In a 1698 letter, Leibniz explained the metaphysical point of view underlying his correlative thinking:

Since all minds are unities, one can say that God is the primitive unity, expressed by all the others according to their capacity (*portee*). His goodness has moved him to act and there are three primordialities in him—power, knowledge, and will. From these there results the operation or creature which is varied according to different combinations of unity and zero.⁵¹

Thus, Leibniz aligned himself with the Augustinian theology of evil as privation of being. Furthermore, he was admittedly following the scholastic idea that essences reflect aspects of the final telos, God. But Leibniz

also sought a rigorous correlation of thought to nature, which scholastic science lacked. He looked for a priori confirmation of the possibility of a universal characteristic even in correlations among mythical, metaphysical, and scientific ideas.

Leibniz took the correlations between the binary system and certain religious numerology as evidence of the likeness of the binary system to a universal characteristic. He commented to Bouvet that, as in *Genesis*, it is sufficient for the creation of the world that it take place merely out of a primary void (0) and God (1), who makes, by the second day, heaven and earth (10). By the seventh (111) day creation is fulfilled.

[T]his is why the last (day) is the most perfect and the Sabbath, for everything on that day is found to be made and full. Thus, 7 is written by 111 without 0. And it is only in this manner of writing that the perfection of the septenary, which is considered sacred, is seen. And it is even more remarkable that its character has some relation to the Trinity.⁴²

The binary system shows that a One is the sufficient reason for all things. Leibniz even surmised, in Figurist fashion, that Fu Xi had considered *Genesis* in designing the Diagram. Thus, for Leibniz, the fact that the binary system corresponds to Biblical numerology (and to his own basic metaphysical ideas of the ultimate origination of things) supported his view that his system corresponded to the world.

Hence, Leibniz' view that the double geometric progression was a real symbol of God's creation shows that correlative thinking was neither alien nor unimportant to him. But if Leibniz maintained an interest in this kind of correlative thinking, Shao thoroughly indulged himself in it. His sacred text was a short passage from the *Xici chuan* of the *Yijing*: "In the Changes there is *taiji*. It produces the two forces." The two forces produce the eight trigrams. Shao interpreted this passage as follows:

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Each of the myriad things embodies the order of the *taiji*, the two forces, the four images, and the eight trigrams.... Each of the trigrams has a nature and substance, but none can be separated from its connection to *qian* and *kun*. The myriad things receive their nature from heaven, but each has its own nature.⁴³

乾 坤

Taking the lead from ancient tradition, Shao explained this basic starting point in his cosmological system:

Yang interacts with yin, and yin interacts with Yang, producing the four images of heaven. The hard interacts with the soft and the soft interacts with the hard, producing the four images of earth. Thus, the eight trigrams are established and, when they are layered together, the myriad things are subsequently produced from them.⁴⁴

The eight trigrams—heaven, water, fire, thunder, wind, lake, mountain, and earth—are the elements of all things. They are generated from the

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interaction of yin (--) and yang (—), which first produce the four images (*xiang*): yang (==), yin (==), the hard (*gang*, ==), and the soft (*rou*, ==). These give rise to the four forms of heaven (sun, moon, stars, and zodiacal space). The hard and the soft produce the four forms of earth (water, fire, ground, and stone), which Shao also called the four substances (*siti*).

Shao also correlated many phenomena in groups of four. As heaven also had four seasons, earth four directions, and people four limbs,⁵⁵ Shao drew correlations to the four virtues (benevolence, propriety, righteousness, and wisdom) and the four kinds of ruling: magically to transform the people, to educate them, to adapt them, and to lead them. These depict the growth and gradual decline in the people's capacity to be ruled and the ruler's capacity to rule. Similarly, natural change undergoes four stages: birth, growth, maturity, and death. Likewise, Shao followed Indian notions of time and set up a cosmic cycle (*yuan*), divided into four subcycles of gradual cosmic decay.⁵⁶

These brief examples scratch the surface of Shao's elaborately correlative thinking, which was central to his system.⁵⁷ The correlations Leibniz drew were not nearly as imposing. However, both thinkers took correlations between sacred cosmogonical ideas and the numbers of the double geometric progression as evidence for the reliability of that progression as a tool for describing nature. Leibniz held that the description of natural change was possible because every created thing had a sufficient reason for its existence. Likewise, Shao based his cosmological system on the primordial *taiji*. In these ways, Leibniz' and Shao's views are similar.

As Leibniz partook of some correlative thinking with regard to the binary system, Shao attempted some rudimentary scientific explanation with it. Even though Shao's cosmology did not approach the rigor of early-modern European science, his explanations of natural change, like those of Leibniz, utilized the periodic series in the binary progression.

An example will show how Shao used the Diagram to explain natural events. Each of the eight trigrams (see Plate I) had a different character, since it resulted from a unique permutation of one of the four images and either a yang or a yin line (see Figure 1). Thus, each reacted differently in combination (i.e., in hexagrams).⁵⁸ The inherent natures of the eight basic natural elements, as well as the ways in which they react to one another is determined. Therefore, natural events are explained accordingly. In the "Dialogue between a Fisherman and a Woodcutter," the following exchange occurs:

"All substantial things can be burned."

"Does water have substance?" asked the woodcutter.

"Yes."

"Can fire burn water?"

The fisherman replied, "The nature of fire can accept it, but cannot continue, so it is extinguished. The substance of water can continue but cannot receive fire, so it is heated. Therefore, there is hot water but not cold fire. It is called mutual bringing-to-rest (*xiangxi*)."⁵⁹

相息

This is an instance of the explanation of natural forces based on a view of the nature of elements as determined by their numerical formulation, that is, their combination of yin and yang lines and their place in the progression of the Diagram. Fire (☲) and water (☵) are on opposite sides of the Diagram of trigrams. They have exactly opposing yin and yang primitives. But this explanation is empty, not shedding much light on why fire is extinguished. All of the possible permutations of the two basic forces in three places are given in the eight (2³) trigrams. Shao infers that the trigrams must represent the eight basic elements. And since the Fu Xi order exhibits the trigrams' ultimate origination, it also fixes a priori (*xiantian*) their natures and the ways in which they must interact. Shao's correlation of the eight trigrams with eight elements was not deduced analytically or empirically but was supported by analogy to further correlations. His cosmology was a complicated system of fours and eights in which the natural kinds and the human faculties all played harmonious roles.⁶⁰

先天

Leibniz thought that everything had a characteristic number.⁶¹ To be sure, the concept of quantitative refinement of explanations of natural force is limited in Shao. He does little to tell us why each particular element (*benti*) is fittingly represented by that particular trigram rather than one of the seven other permutations. And Shao's use of quantitative measurement was also limited. However, the following exchange shows his understanding of metric refinement:

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The woodcutter said to the fisherman, "I have carried firewood and lifted one hundred *jin* without any strain to my body. But if I add ten *jin*, I strain. If I may ask, Why?"

The fisherman said, "... When it comes to fish and firewood, there is a difference. But the desire and strain for them are the same. Whether it's one for whom one hundred *jin* is beyond his strength or one for whom ten *jin* is beyond his strength, the amount beyond one's strength, even if it were one speck, would be harmful. How much more would ten *jin*!"⁶²

斤

Shao understood the universal applicability of quantitative measurement. He at least understood force as a primitive that can be found in varying degrees in things and events. However, Shao did not develop this idea in the direction of advocating rigorous science.

Rather, Shao relied mostly on the verification provided by correlative thinking. Explanation of phenomena was a matter of extrapolating on the first eight permutations of the double geometric progression. As we have seen, the four heavenly bodies are constituted by yin and yang in per-

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mutation. And the four earthly substances were constituted by the other two images, the hard and the soft (which themselves emerged from the first two images, yin and yang). Strictly speaking, these were not the eight trigrams themselves, which emerged from the permutation of the four images (the first four permutations of yin and yang, given in the second row in Figure 1) as heaven, earth, water, mountain, wind, fire, lake, and thunder. Shao considered the heavenly and earthly bodies to be forms (*xing*) and not the elements of things (the trigrams). He explains natural phenomena with them all:

The sun causes heat, the moon causes cold, the stars cause light, and the zodiacal spaces cause night. Night, heat, cold and light together exhaust the transformations of heaven. Water makes rain, fire makes wind, ground makes dew, and stone makes thunder. Rain, wind, dew and thunder together exhaust the changes of the land. Heat transforms the nature of things, cold transforms the feeling of things, light transforms the forms of things, and night transforms the substance of things. Substance, nature, feeling, and form together exhaust the influences of moving on stationary [things]. Rain changes walking things, wind changes the flying things, dew changes grassiness of things, and thunder changes woodiness of things. Walking, flight, grassiness, and woodiness together exhaust the responses of things.⁶³

Material force—*qi*—is subject to transformation among the elementary configurations of the eight trigrams. The causation observed in nature is evidence for the validity of the extrapolations made in Shao's cosmology. Thunder has great impact on the creation of wood, and wind effects birds. Thus, *qi* changes in ways predicted by the sixty-four permutations of the trigrams. "One can observe heaven [i.e., nature] to grasp the story [of nature], or one can examine earth."⁶⁴ So, Shao takes his system to be confirmed by the loose correlations of experience. But the system is nevertheless a priori (*xiantian*) in that it is derived from basic assumptions about the unobserved origin of the world.⁶⁵ Shao assumed that the *xiantian* Diagram depicted the origin of things as "both temporally and logically prior"⁶⁶ to the world.

While these notions did not quite amount to the concepts of scientific hypothesis or theory verification and thus were only proto-scientific, Shao did make an epistemological argument concerning these correlations. For Shao, the harmony (*he*) of the various natural correlations in fours and eights gave credence to the idea of the objectivity of four sense faculties, for these neatly matched their four modal objects (sight, sound, smell, and taste). These eight faculties and objects also exhibited interesting correlations to the eight elements.⁶⁷ Shao explains the matter in the "Dialogue between a Fisherman and a Woodcutter":

The woodcutter said to the fisherman, "That by which people can be conscious of the myriad things—how do you know it is true?"

The fisherman said, "We say that one's eyes can receive the color of the myriad things. Ears can receive the sound of the myriad things. The nose can receive the smell of the myriad things..." (sec. 15)

"[To observe things] is not to observe them with the eyes but to observe them with the mind. It is not to observe them with the mind but to observe them with principle. There is nothing under heaven that does not have principle, nothing that has no nature, and nothing that has no destiny. That which we take to be called principle can be known after it is investigated.

"That by which a sage is able to unite with the particularities of the numerous things is called his ability to observe objectively, ... not to observe things with the self [but] to observe things with things." (sec. 5)

"... Also, we refer to [the sage's] ability to substitute for heaven's thoughts with the mind, to substitute for heaven's speech with the mouth..." (sec. 15)

Shao gives indication of an epistemological system in which objective truth, an absolute conception of the world, is in principle available to the rigorous inquirer and is reached by considering rational principles (*li*) rather than by succumbing to our biased opinions of things. The senses are apparently sources of knowledge, since they are perfectly coincident with their respective modes. But again, the mind must overcome its tendency to twist the data toward subjective conclusions.

One basic a priori belief from which Shao deduced that the numerical series in the double geometric progression reflected principles of natural change was given in the *Xici chuan*: "Heaven is one, earth two, heaven three, earth four, heaven five, earth six, heaven seven, earth eight, heaven nine, and earth ten."⁶⁸ The text shows that the eight elements have orthodox legitimacy in the *Yijing*. Shao holds that five and ten "are the poles of being and nonbeing,"⁶⁹ and, thus, are not part of creation. This leaves eight numbers—four yang and four yin—corresponding to the trigrams of the Diagram.

Shao finds further support for his Diagram in an ancient mathematical diagram, the *He Tu* (River Diagram), a derivative of the ancient magic square, the *Luo Shu*.⁷⁰ As Figure 2a shows, the *Luo Shu* sums to fifteen when any three of its numbers in one row, column, or diagonal are added together. The reasons for the adjustments which the *He Tu* (Figure 2b) represents need not concern us here.⁷¹ As Cammann has shown, when the even numbers are connected, the probable origin of the *taiji* symbol is evident (Figure 2c).⁷²

The Diagram is faithful to the sacred alternation depicted in the *He Tu* and the *Xici chuan*. Starting from the top trigram (*qian*) and proceeding counterclockwise until *zhen* (at 7:30), and then proceeding from *sun* (at 1:30) clockwise to earth, Shao counts out the eight trigrams as they are generated in the binary permutation of yin and yang (see Figure 1).⁷³ Thus, the Diagram reflected, for Shao, the numerology of the *He Tu*.⁷⁴

Hence, Shao argued, the Diagram's validity followed from certain

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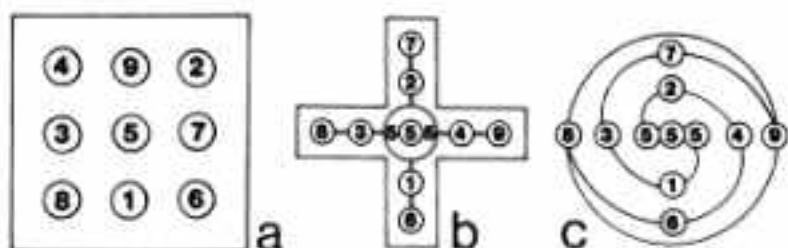


Fig. 2. a, the Luo Shu, the ancient magic square; b, the He Tu (River Diagram), a derivation of the Luo Shu; c, origins of the taiji symbol. From Schuyler Cammann, "Some Early Chinese Symbols of Duality," *History of Religions* 24, c 1985, University of Chicago Press. Reprinted with permission.

fundamental beliefs about nature that had been held to be confirmed by a long tradition of correlative evidence. The Diagram represented the mechanics of the basic metaphysical view of *taiji*; it explained the nuts and bolts of what the *taiji* symbol only represented in sweeping outline. Although the Diagram was derived mechanically from repeated divisions, it was the sequence of numbers going around the Diagram that was so important.

Shao had a keen interest in the numerical series, as his doctrine of
 消長 (waning and waxing (*xiaozhang*)) shows. The Diagram portrays the gradual and orderly filling-in of yang lines (from the bottom hexagram [䷊] clockwise to the top) and the similar elimination of yang lines (proceeding from the top [䷊] clockwise to the bottom):

The ascent is waxing and the end is waning. That is why yang [i.e., life, growth, etc.] arises at the bottom and yin arises at the top.⁷⁵

Shao noted the repetitive series of the double geometric progression in the trigrams and hexagrams:

The woodcutter said to the fisherman, "The four images produce the eight trigrams. What are the eight trigrams called?"

"They are called heaven, earth, fire, water, lake, mountain, thunder, and wind. They alternate between the characteristics of flourishing and declining, and ending and beginning. Thus, when they are layered, the sixty-four [hexagrams] arise from this, and the way of the Changes is set up."⁷⁶

In the Diagram we see yin and yang alternately flourishing and declining, just as we see ones gradually replace the zeros in the binary number system. Since the order in the Diagram emerges from the basic assumption that creation is driven by yin and yang, the structure of the waxing and waning of all things is determined by this creation. Therefore, this structure is knowable a priori, for Shao. Thus, there is an observational component in Shao's view of waxing and waning:

Waning and waxing have to do with seasons... If the sages didn't know the way to follow the evil and prosperity of the seasons, how could they know how changes are affected?... That which drives waxing and waning is change.⁷⁷

Not only natural phenomena, but also historical periods are the subjects of waxing and waning:

If one investigates [the way of heaven and human affairs], the historical periods can be seen. This is the principle of waxing and waning.⁷⁸

The chief result of Shao's interest in the periodic series of the Diagram was the hallmark of his system: his calculation of the cyclical history of the world. Shao calculated the duration of one cycle (*yuan*), twelve epochs, as 129,600 years. Like each month, each epoch was marked by the incremental progress of yang (or yin) with a change of a line that hadn't been changed before in that cycle. For example, the second, third, and fourth epochs and months were represented by the following hexagrams: ☱, ☲, ☳. (In Plate I the hexagrams representing the twelve months [and twelve epochs] are indicated by arrows.)

This is as close as Shao came to recognizing the place-values in the binary system. He never explained the fact that these equal periods of time are represented by hexagrams that are not evenly spaced apart. The spaces between the hexagrams in question increase geometrically, but, since they represent equal periods of time (months, for example), they presumably ought to be evenly spaced around the circle in the Diagram. Shao never spoke to this discrepancy.⁷⁹ Furthermore, even though Shao was adept at arithmetic, he never recognized the squares and cubes in the series. But the point is that Shao saw the series as an important cosmological clue, rather than merely deriving the hexagrams by dumb, mechanically repeated division.

Instead of quantitative and verificational experimentation, Shao emphasized a religious perspective on the Diagram. He says:

The study of *xiantian* is the law of the mind. Therefore, the diagrams all arise from within oneself. The myriad changes and the myriad affairs are produced in the mind. Although the diagrams lack texts, on any day I speak of them and have never been separated from them. For the principles of heaven and the myriad things lie in them.⁸⁰

Insofar as Shao was, as we will see below, primarily interested in religious knowledge, the case for dismissing the *Yijing* / Binary System Episode as a coincidence is somewhat justified. Yet Leibniz, himself, though an inventor, was no experimental scientist. And he, too, based his views on what he took to be an a priori religious perspective. And Shao was en-

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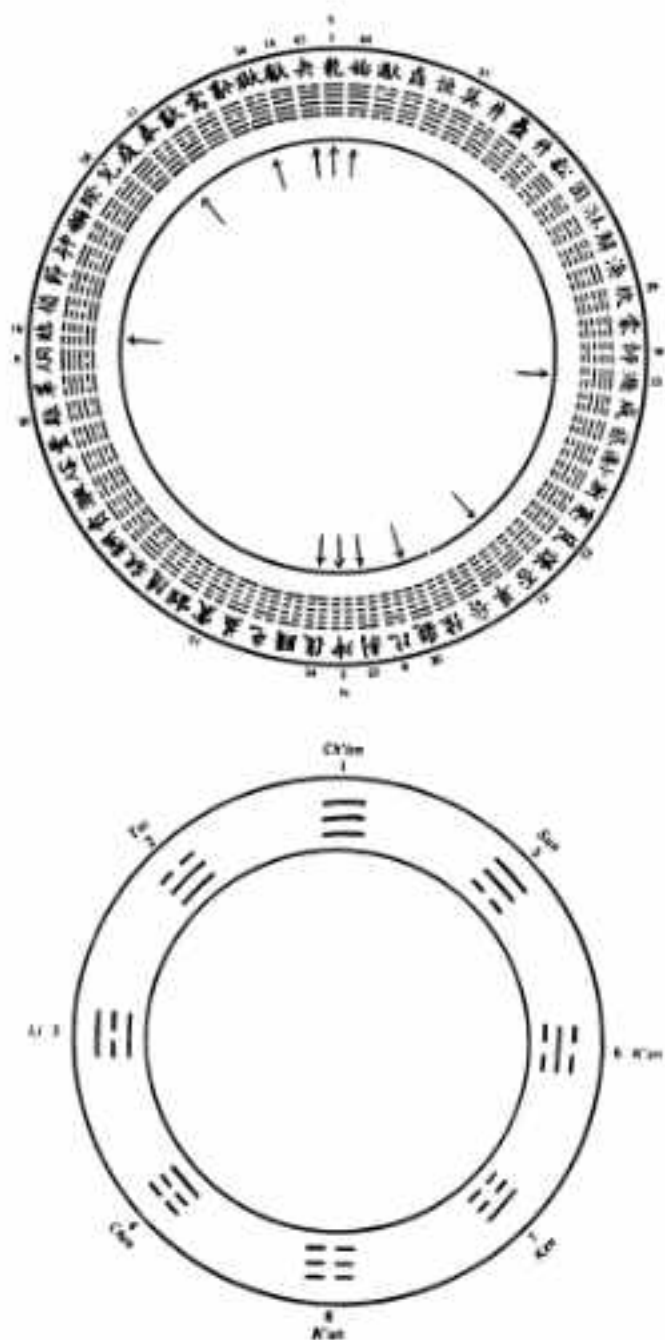


Plate 1. The Diagram: Shao's calculation of the cyclical history of the world. Arrows indicate the twelve months (and twelve epochs). From Yu-Lan Fung, *History of Chinese Philosophy*, Vol. 2, Princeton University Press, 1953. Reprinted with Permission.

gaged in something similar to Leibniz' ideas of conjecture a priori and conjecture a posteriori. Of course, he never conceived of variety and simplicity as the principles of conjecture a priori, but he did take as basic to his proto-scientific activities a metaphysical view of the cosmos as a harmonious system whose parts mutually reflect each other in virtue of their common binary origin and structure. In this Shao was like Leibniz, who also thought that, as Lovejoy puts it, "we can discover factual truths of science by tracing out the implications of the fundamental scheme of values which the universe expresses."⁸¹

Shao made analogical a posteriori inferences. But his purpose was not to verify these conjectures, since he apparently never seems to have changed his principles for any observation that contradicted them. For example, his *xiantian* year is 360 days, even though he must have known that one millennium earlier Chinese astronomers had calculated the year to 365 1/4.⁸² Thus, Shao's a priori cosmology was intended to speak to the primordial nature of things, and not to their phenomenal circumstances, which are a degenerate version of that nature. In a sense, Shao's science was about the best of all possible worlds; but that world, for him, no longer existed. Thus, he kept a priori and a posteriori methods separate, instead of looking to their interplay for clues about verifying hypotheses. Yet Leibniz would have agreed that conjecture a priori was formally separable from conjecture a posteriori. He says:

... do we not say universally that fire ... usually flares up and burns when wood is kindled, even if no one has examined all such fires, because we have found it to be so in those cases we have examined? That is, we infer from them and believe with moral certainty ... but this moral certainty is not based on induction alone and cannot be wrested from it by main force but only by the addition or support of ... universal propositions which do not depend on induction but on a universal idea...⁸³

This universal idea was that the simplest concept would yield the widest explanatory power. It played a crucial role in induction by supporting the notions that similar causes produce similar effects and that the existence of undetected causes ought not to be assumed.

Shao's system was not rigorous science in the early-modern sense, for the a priori and observational components were not subjected to the methodological revision to which Leibniz and his contemporaries put Western science. As we have seen, Shao's explanation of fire being extinguished by water is empty. And he certainly lacks the kind of inference *more geometrico* that Leibniz rightly held would preserve truth. He provides no reasons to support his inferences about the natures of the trigrams. For example, why should thunder be represented as it is ($\equiv$)? Shao's a priori principles were not as scientifically sophisticated as Leibniz'. Yet they were coherent principles that relied on an idea of

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the perfect harmony of things. In this sense, Shao's system was proto-scientific.

But instead of evaluating Shao's thought purely in terms of its meeting Western scientific standards, we need to understand him on his own terms. In order to explain the significance of the *Yijing* / Binary System Episode completely, then, we must understand why Shao chose to go only as far as this proto-scientific cosmology. This involves explaining the different goals of Shao's philosophy, for these account for Shao's thinking along lines other than those followed by the European founders of early-modern science.

Shao on Inquiry and its Goals

Shao's views about the role of quantitative and mechanistic thinking explain, to some extent, why he placed certain limits on scientific thinking and did not develop the binary system in the direction that Leibniz did. Shao tried to avoid this scientific direction and provided reasons for its inconsistency with his system. He took Buddhist, Taoist, and Confucian principles of self-cultivation as antithetical to scientific observation. I have laid out Leibniz' rigorously scientific development of the double geometric progression and Shao's proto-scientific developments of it. Now a brief account of Shao's positions on knowledge and science will distinguish the philosophical goals for the sake of which Leibniz and Shao refined the double geometric progression in their different ways. Then we will be able to see how the Diagram bore more than a formal similarity to the binary system, even though the Diagram was not used as a number system.

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Shao's theory of knowledge was directed toward defining objective observation (*fangan*), but it differed from Western accounts in its heavily Buddhist and Taoist influences. For Shao, a necessary condition for objective observation was the overcoming of the self's egocentric point of view, for the self tends to take things as real which are not real. Using Chinese Buddhist terminology, such as no-mind (*wuxin*) and no-idea (*wuyi*), Shao described objectivity as seeing things from the things' own point of view: "... one can reach the myriad things through no-mind ... 'no-mind' means 'no-idea'. No-ideas don't 'self' things. If you don't self things, then you can 'thing' things."⁸⁴ Shao meant that objective talk about nature requires getting rid of one's individual and subjective point of view: "If one speaks following the season or speaks in correspondence to change, then [this] speech is not in one's self."⁸⁵

Shao explains the way in which our reason is subject to distortion by the passions. "Indulging the self leads to feelings. Feelings lead to obscurity. Obscurity leads to confusion. Taking the lead from things leads to nature. Nature leads to spirit. Spirit leads to enlightenment."⁸⁶ Leibniz would have agreed with this view. He says, "The denial of self is the

hatred of non-being which is in us and the love of the origin of our self-being, or God."⁸⁷ But Shao inferred consequences that Leibniz would not have accepted. Shao says, "The mind is *taiji*. The human mind should be like still water. Then it will be fixed. If it is fixed, it is tranquil. And if it is tranquil, it is enlightened."⁸⁸

Thus, Chinese meditative traditions had influence on Shao's epistemology. Rather than attributing knowledge to rigorously inferential thinking, as Leibniz did, Shao attributed it to a degree of mental training in which individual impulses are suppressed so that the mind can reflect truth without distortions: "If the mind is one and not divided, then it can correspond to the myriad things. This is why the superior person empties his mind and doesn't move."⁸⁹

Furthermore, Shao's epistemology was also shaped by Confucian notions of moral cultivation. "*Xiantian* studies depend upon sincerity."⁹⁰ For example, the passage on lifting weights is not a precursor to Shao's development of rigorous measurement techniques. On the contrary, the moral of the passage was given by the fisherman: "From now on I know that one who takes stock of one's strength and then moves is wise."⁹¹ This was an allusion to Shao's doctrine that we should seek to fulfill our individually given capacities without an egoistic desire for greater glory than our fates in fact hold for us. This is the upshot of Shao's excursus into statics!

Also, Leibniz and Shao differed in their views of the role of knowledge in history. Shao thought that history was cyclical, eternally returning to the same. For Leibniz, on the other hand, "Progress never comes to an end."⁹² He thought that a thing is happy

to the degree that its power is greater over the things that surround it, and its suffering from external things is less. Hence, since the power proper to the mind is understanding, . . . only to the extent that our reasonings are right are we free, and exempt from the passions which are impressed upon us by surrounding bodies.⁹³

For Shao, fate held no possibility of progress beyond the inevitable waxing and waning of order. Happiness involved contentment with, not dominion over, nature. Empirical science seemed to him a wrongheaded and egoistic effort to control nature. Shao held that scientific thinking leads one astray from the only kind of thinking that can discover the truth about nature. In particular, Shao put China's astronomy (*huntian*) in the same category as divination—mere "technique" (*shu*)—since both trends agreed on the premise that the individual could gain control of his place in nature through personal effort with special numerical arts:

Phenomena, whether great or small, all have the principle of heaven and human beings. To cultivate oneself is up to people. Whether one meets or

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does not meet [good fortune] is up to heaven. Not to disturb the mind when encountering misfortune is how one obeys heaven. To take with force, to have to obtain, is to oppose the principle of heaven.⁹⁴

One way of trying to take knowledge by force was the manipulation of numbers in divination. "Average people use numbers to enter into techniques. Therefore, they don't enter into principles."⁹⁵ Here, "techniques" refers to divination, which investigates natural principles for selfish gain. Leibniz would have agreed with Shao's caution toward "arts of divination, which are nothing but cheating."⁹⁶ However, Shao places science into the same category:⁹⁷

The images and numbers of heaven can be obtained and calculated, but the working of its spirit cannot be obtained and measured. Heaven can be exhaustively explained with principle, but it cannot be exhaustively explained with forms. How can the techniques of astronomy exhaustively explain heaven by means of forms?⁹⁸

Astronomy in China was more than one thousand years old by Shao's time, and it represented a respectable body of measured observations of the heavens. Astronomers were working on a model of the cosmos shaped like a spherical egg, with the earth as the yolk and the constellations on the surrounding shell.⁹⁹ But the universe could not be perfectly represented by a model, for Shao.¹⁰⁰ Models were subjective and forced representations: "If there is something one doesn't understand, one can't understand it by force. If one understands by force, then there is a self. If there is a self, then one loses principle, and enters into technique."¹⁰¹ According to Shao, science, like divination, manipulates nature with sophisticated techniques, and, thus, it is selfish and cannot yield objective knowledge. Furthermore, the kind of thinking involved with these inauthentic and insincere techniques even tends to hide the way to real knowledge.

Shao provides a metaphysical argument to support these strictures on rigorous science. The spirit (*shen*) of nature is the moving aspect of the *taiji*.¹⁰² Spirit produces numbers, from which all things come. As we have seen, "the workings of spirit cannot be obtained and measured."¹⁰³ This is because spirit is nonspatial (*wufang*).¹⁰⁴ "If it were confined to one location, then it could not change and transform, so it wouldn't be spirit."¹⁰⁵ Not being spatial, then, spirit can't be divided.¹⁰⁶ Thus, it can't be measured. For this reason, rigorous science is futile, if it is intended to explain the world completely.

Leibniz, refuting the Cartesians, argued that force transcended geometrical representation. Shao seems to have argued similarly. But Shao took this conclusion as evidence that a scientific inquiry into the spirit of nature would be fruitless. Whereas Leibniz tried to strike a chord between final causes and mechanical processes in his physics, Shao argued

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that spirit eludes scientific representations, since, as the cause of action, it can't be still or in one space. "There is no place that spirit is, and no place that it is not."¹⁰⁶ Moreover, Shao claimed that science is on the epistemic level of common divination, since it uses models and hypotheses and is, therefore, a subjective creation.

Conclusion

That Shao Yong placed such caveats on his philosophy of science shows that, for spiritual and epistemological reasons, Shao thought that technical uses of the Diagram ought to be limited. Hence, understandably, Shao would not have endeavored to develop a number system with the Diagram. But it does not show, against the other evidence, that the meanings of the Diagram and the binary number system were worlds apart. For the other evidence shows that the role of the Diagram in Shao's philosophy bore significant, and not purely formal, similarity to Leibniz' binary system.

Shao's system lacked vital features of science, such as falsifiability.

Table 2. Leibniz and Shao on Science

	Leibniz	Shao
Theory Formulation	A prior and a posteriori conjecture.	Correlative thinking and observation crudely analogous to Leibniz' conjecture.
Cataloguing of Natural Kinds	Natural kinds are analyzable down to their elements through observation of their mutual interactions.	Same as Leibniz
Metaphysical Underpinnings	"Mechanical laws are finally resolved into metaphysical reasons." The binary system explains natural phenomena since it reflects the metaphysics of creation.	Same as Leibniz
Falsifiability	Verification is the essence of conjecture a posteriori. Laws are deduced in consultation with observation.	(None; Fu Xi had already done the observational work.)
Goals	1. To develop a natural language. 2. To develop technology to meet (particularly medical) needs.	1. To recognize the principles of natural change in order to align one's spirit with them.

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While it might be considered egocentric for a scientist not to change his hypothesis even when the evidence has falsified it, Shao left falsifiability out of his picture of the proper explanation of nature with the Diagram. For Shao, the objective modeling of particular phenomena represented a selfish desire to control phenomena technologically, while, after all, phenomena were only a degenerate version of the original creation of things, whose natures were laid out in the Diagram.

Nevertheless, the features of Shao's system add up to a proto-science and thus support the view that Leibniz was not completely wrong when he claimed that the Diagram was scientific (see Table 2). Both Shao and Leibniz thought that "mechanical laws are finally resolved into metaphysical reasons," reasons which reflect the creation of the world in a binary, double geometric progression whose periodic series depict natural change and natural kinds. Both Shao and Leibniz thought that the principles in the double geometric progression helped to categorize the elements and the natural kinds. And both thought that the human mind played a central role in the world by utilizing reason and sensory input. Thus, knowledge was meant to serve the human spirit and, as Leibniz emphasized, the human body.

Had Leibniz been able to study Shao Yong, he would have found that the strong version of his theoretical claim that the Diagram was scientific was false. Shao's system amounts only to a proto-science in which the Diagram is not a number system. But Leibniz nevertheless would have encountered a like mind. He would have found a philosopher interested in pushing a view of nature that was binary and mechanistic but which also embraced the orthodox teleological views of his particular culture.

NOTES

Special thanks to David Blumenfeld and Alan Berkowitz for their guidance, and to two referees for their patient and painstaking devotedness to inquiry. Thanks also to Mark Woodhouse and Steve Prothero for their helpful advice.

- 1 – Wilhelm finds the episode baffling and in need of "clearing up." See Helmut Wilhelm, *Change: Eight Lectures on the I Ching* (New York: Harper and Row, 1960), pp. 90f.

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- 2 – The *Fu Xi Liushisigua tu*, depicted in my Plate I. I will refer to it as the Diagram.

- 3 – Alan Berkowitz, "On Shao Yong's Dates," *Chinese Literature: Essays, Reviews* 5 (1983): 91–94.

- 4 – Leibniz explains his understanding of the Diagram as a binary system primarily in two writings: “The Leibniz-Bouvet Correspondence” (1697–1707) (*Leibnizbriefe*, 105, Niedersächsische Landesbibliothek, Hanover, Germany), and the *Discourse on the Natural Theology of the Chinese* (1716; hereafter referred to as *DNTC*). The latter shows Leibniz’ admirable understanding of various aspects of Chinese philosophy but contains nothing on the topic of the *Yijing* not found in the correspondence. So my remarks here will focus on that correspondence.
- 5 – Another aspect of Leibniz’ universalism was his involvement in the church unification project, the effort to reunite the Protestant and Catholic churches. See E. J. Alton, *Leibniz: A Biography* (Boston: Hilger, 1985), pp. 180f.
- 6 – In the Preface to his *Novissima Sinica* (1699), Leibniz praised Confucian ethics as being more advanced than Western ethics (Donald Lach, trans., *The Preface to Leibniz’ Novissima Sinica* [Honolulu: University of Hawai’i Press, 1957]).
- 7 – For a discussion of Bouvet, see David E. Mungello, *Leibniz and Confucianism: The Search for Accord* (Honolulu: University of Hawai’i Press, 1977), pp. 36–46, and David E. Mungello, *Curious Land: Jesuit Accommodation and the Origins of Sinology*, *Studia Leibnitiana: Supplementa*, vol. 25 (Stuttgart: Steiner-Verlag-Wiesbaden, 1985), pp. 300f.
- 8 – Figurism was the view that all cultures spring from the ancient Judeo-Christian line and that their pre-Christian histories contained intimations of the future Christian teachings. See Mungello, *Curious Land*, pp. 307–328.
- 9 – “Leibniz-Bouvet Correspondence” (hereafter referred to as *L-B*), p. 184 (28 February 1698). I cite from the manuscript of the unpublished translation by Alan Berkowitz and Thatcher Deane.
- 10 – *L-B*, pp. 184 (28 February 1698), 216–217 (4 November 1701).
- 11 – *Ibid.*, p. 242 (1703?).
- 12 – *Ibid.*, p. 249 (1703?).
- 13 – *Ibid.*, p. 175 (2 December 1697).
- 14 – *Ibid.*, p. 176. Against the Cartesians, who argued that motion (the product of mass and velocity) is conserved, Leibniz argued that it is force—“*vis viva*” (the product of the mass and velocity-squared)—that is conserved. See “A Brief Demonstration of a Notable Error of Descartes and others Concerning a Natural Law” (1686), in Gottfried Wilhelm Leibniz, *Philosophical Papers and* James A. Ryan

Letters, trans. Leroy E. Loemker (Dordrecht: D. Reidel, 1970), pp. 296–302 (hereafter referred to as *L*).

- 15 – *L-B*, p. 175 (2 December 1697). Leibniz said in the letter that he found “that the [Aristotelian] philosophy . . . is sound . . . and that it is necessary to consider in nature not only matter, but also force, . . . the forms of the ancients, or entelechies.”
- 16 – There are similarities between *li* and the monads. Both terms represent noumena and indicate the rational unity or “fit” of things in nature. Like the Chinese, Leibniz was organicist in his thinking. Thus, both the monads and the *li* of individual things reflect the situation in which they are placed. But while there is no indication that particular *li* are causally independent of each other, each monad plays out only its internal rationality and is not effected by external causes. There is little sense of mechanistic necessity in *li*. For one can deviate (usually with disastrous results) from one’s *li*, but deviation from one’s monad, or what is contained in one’s concept, is impossible. See Mungello’s discussion in *Leibniz and Confucianism*, pp. 79ff. Leibniz also equates *li* with the Aristotelian formal cause “entelechy” (*DNTC*, p. 76).
- 17 – *L-B*, p. 172.
- 18 – “On the General Characteristic,” in *L*, pp. 221–228.
- 19 – While a nominal definition serves only to distinguish a thing from all others, a real definition explains the elements of a thing and the reasons why these elements can in fact combine to produce the distinctive qualities named in the nominal definition. See “On the Universal Synthesis and Analysis, or the Art of Discovery and Judgement” (1679), in *L*, pp. 230–231.
- 20 – “Two Studies in the Logical Calculus,” in *L*, pp. 235–247.
- 21 – *L-B*, p. 170 (2 December 1697).
- 22 – Leibniz asked Bouvet about the contemporary Chinese interpretation of the Diagram. See *L-B*, p. 205 (15 February 1701).
- 23 – *L-B*, p. 204 (15 February 1701), and pp. 249–250 (1703?).
- 24 – *DNTC*, p. 158.
- 25 – Note that the term used during Leibniz’ time for what we think of today as natural science was “natural philosophy.” “Science” was a broader term referring to systematic inquiry in general.
- 26 – See Don J. Wyatt, “Chu Hsi’s Critique of Shao Yong: One Instance of the Stand against Fatalism,” *Harvard Journal of Asiatic Studies* 45 (1985): 649–660.

- 27 – “On the Elements of Natural Science” (1682–1684), in *L*, p. 283.
- 28 – “On Freedom” (1679), in *L*, pp. 263–266.
- 29 – There is some question as to whether the most perfect possible world must involve a compromise between the two goals of variety and simplicity or whether, on the contrary, the world with the most variety must have the simplest laws, such that that world has more variety than any other and more simplicity than any other. The former reading is explained by Nicholas Rescher in *Leibniz’ Metaphysics of Nature: A Group of Essays* (Dordrecht: Reidel, 1981), pp. 8f. The latter reading is argued by David Blumenfeld in “Perfection and Happiness in the Best Possible World,” in *The Cambridge Companion to Leibniz*, Nicholas Jolley, ed. (Cambridge: Cambridge University Press).
- 30 – See Rescher, *Leibniz’ Metaphysics of Nature*, pp. 40f.
- 31 – *Discourse on Metaphysics*, no. 22, in *L*, p. 317.
- 32 – Leibniz distinguishes between moral certainty and metaphysical certainty. The former refers to robustly and experientially confirmed beliefs. Metaphysical certainty attends those beliefs (such as the law on noncontradiction) the denial of which is inconceivable. See Leibniz, “On the Method of Distinguishing Real from Imaginary Phenomena” (1690), in *L*, p. 364.
- 33 – The truths of arithmetic are deductively certain, because they are reducible to identities. The mathematical inspiration for the rules for forming scientific hypotheses lay in the infinitesimal Calculus, in which identities are only infinitely closely approached. See Rescher, *Leibniz’ Metaphysics of Nature*, pp. 48f.
- 34 – See *Tentamen Anagoricum* (1696), in *L*, pp. 477–485.
- 35 – “On a Method of Arriving at a True Analysis of Bodies and the Causes of Natural Things” (1677), in *L*, p. 173.
- 36 – “On the General Characteristic” (1679), in *L*, p. 221. These ideas reflect Leibniz’ relatively early thinking.
- 37 – “Letter to Walter Von Tschirnhaus” (1678), in *L*, p. 193.
- 38 – Mungello, *Curious Land*, p. 195. See also pp. 174–207, where Mungello traces the history of the efforts to find a universal language.
- 39 – *Ibid.*, p. 195. It is not clear that Mungello is referring to computer science applications of Leibniz’ ideas, but I take these to be the principle legacy of Leibniz’ ideas.

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- 40 – "On a Method of Arriving at a True Analysis of Bodies and the Causes of Natural Things" (1677), in *L*, p. 174.
- 41 – "On the Elements of Natural Science" (1682–1684), in *L*, p. 282.
- 42 – *L*, p. 79.
- 43 – "On the Method of Arriving at a True Analysis of Bodies and the Causes of Natural Things," (1677), in *L*, p. 174.
- 44 – See Schuyler Cammann, "Some Early Chinese Symbols of Duality," *History of Religions* 24 (1985): 215–254, and Mungello, *Leibniz and Confucianism*, pp. 64–65.
- 皇極經世書 45 – Shao Yong, *Huangji jingshi shu* (Sibu beiyao ed.), 7A.24b (hereafter referred to as *HJS*). All translations of Shao given here are my own.
- 46 – Mungello allows that this reversal would "not invalidate the correspondence in principle" (Mungello, *Curious Land*, p. 319).
- 47 – E. J. Aiton and E. Shima, "Gorai Kinzo's Study of Leibniz and the *I ching* Hexagrams," *Annals of Science* 38 (1981): 89. See also Aiton, *Leibniz*, p. 247, and Mungello, *Leibniz and Confucianism*, pp. 52, 140.
- 48 – In John Henderson's words, correlative thinking, in general, "draws systematic correspondences among aspects of various orders of reality or realms of the cosmos, such as the human body, the body politic, and the heavenly bodies" (John B. Henderson, *The Development and Decline of Chinese Cosmology* [New York: Columbia University Press, 1984], p. 1). In a lengthy exposition, A. C. Graham describes correlative thinking as the reaching of conclusions by association rather than by analysis; see his *Yin-Yang and the Nature of Correlative Thinking* (Singapore: The Institute of East Asian Philosophies, 1986). See also Anne D. Birdwhistell's discussion, *Transition to Neo-Confucianism: Shao Yung on Knowledge and Symbols of Reality* (Stanford: Stanford University Press, 1989), pp. 63–64. Correlative thinking underlay both Shao's and Leibniz' approaches to metaphysics and cosmology to some extent, and thus blurred the distinction between these two fields of inquiry. Metaphysics examines essences and reality, while cosmology tries to explain the origin and structure of the universe. The two fields tend to dovetail if one, as Shao and Leibniz each do, correlates an account of things' essences with an account of origins. For both Shao and Leibniz, things are essentially determined by their primordial origin in one and nothing, yin and yang, or *taiji* and *qi*.

- 49 – *L-B*, p. 197 (15 February 1701).
- 50 – “Letter to Magnus Wedderkopf” (1671), in *L*, p. 146.
- 51 – From a letter to Morell (May 1698), in G. Grua, ed., *G. W. Leibniz: Textes inédits d’après les manuscrits de la Bibliothèque provinciale d’Hanovre*, 2 vols. (Paris, 1948), p. 126; translation by David Blumenfeld.
- 52 – *L-B*, p. 252 (1703?).
- 53 – *HJJS*, 8B.9b.
- 54 – *Ibid.*, 7A.24b.
- 55 – *Ibid.*, 8B.6b–7a.
- 56 – See, for example, *HJJS*, 6.15a; Birdwhistell, *Transition*, pp. 138 ff., and Fung Yu-lan, *A History of Chinese Philosophy*, 2 vols. (Princeton: Princeton University Press, 1953), vol. 2: 469 ff.
- 57 – See Birdwhistell, *Transition*, pp. 124 ff.
- 58 – These ideas were not original in Shao, but are found in the “Shuo-gua” section of the *Yijing*: “... heaven and earth receive their determinate positions; mountain and [lake influence each other]; thunder and wind excited each other the more, water and fire did each other no harm (Legge, *I Ching*, p. 424).
- 59 – Shao Yong, *Yuqiao wendui*, sec. 1 (hereafter referred to as “Dialogue” and by section number). Though the authorship of the Dialogue is usually attributed to his followers, the ideas expressed in it are consistent with Shao’s views. In many cases the Dialogue merely quotes long passages from Shao’s own *Huangji jingshi shu*. Therefore, I think relying on the Dialogue for evidence of Shao’s views is valid. 漁樵問對
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- 60 – Birdwhistell, *Transition*, pp. 125 f., 241–242.
- 61 – “On the General Characteristic” (1679), in *L*, p. 221.
- 62 – Shao, “Dialogue,” sec. 9.
- 63 – *HJJS*, 5.2a–b.
- 64 – *Ibid.*, 8B.6b.
- 65 – Birdwhistell disagrees, finding the previous passage to be good evidence that “*xiantian*” should not be translated as “*a priori*” (Birdwhistell, *Transition*, p. 85). Both Fung and Chan use “*a priori*.” See Fung, *A History of Chinese Philosophy*, vol. 2: 460 (1953), and Wing-tsit Chan, *A Source Book in Chinese Philosophy*, p. 493 n. 25 (1963) (both Princeton: Princeton University Press). Cammann cautions against confusing widely different uses of the term James A. Ryan

xiantian, since “*xian*” originally referred only to the fact that the Fu Xi diagram was invented historically “prior” to the King Wen diagram (Cammann, “Some Early Chinese Symbols of Duality,” pp. 215–254).

66 – Wyatt, Smith, et al., *Sung Dynasty Uses of the I Ching* (Princeton: Princeton University Press, 1990), p. 112.

67 – Birdwhistell, *Transition*, p. 128.

68 – Legge, *I Ching*, p. 365. Shao quotes this passage in *HJJS*, 7A.15a.

69 – *HJJS*, 7A.6b. Shao also says, “Yin lacks one and yang lacks ten” (*HJJS*, 7A.15a).

70 – Schuyler Cammann, “The Magic Square of Three in Old Chinese Philosophy,” *History of Religions* 1 (1961): 37–80, at p. 58. Cammann, in “Some Early Chinese Symbols of Duality,” dates the *Luo Shu* to the fifth century B.C. (p. 222).

71 – Suffice it to say that the 7 and 2 were switched with the 4 and 9, and a 10 (two fives) was added to the middle. Cammann (“Some Early Chinese Symbols of Duality”) provides a detailed explanation of the evolution of both diagrams.

72 – *Ibid.*, p. 234.

73 – Shao counts *zhen* as female, even though it is the trigram for the “eldest son,” and *sun* as male, even though it is the “eldest daughter.” He might not have had to force this fit if he had realized that the Diagram corresponds to, and was probably derived from, the *Luo Shu*. However, in the second century A.D., a lasting mistake was made, and the Diagram was associated with the *He Tu* (Cammann, “Some Early Chinese Symbols of Duality,” p. 252). As Cammann shows (*ibid.*), the members of the family of trigrams match the *Luo Shu* sequence, which, when fire (*li*) and water (*kan*) are considered to be androgynous (as they traditionally are considered), provides for the following alternation of yin and yang trigrams: *qian* (yang), *dui* (yin), *li* (yin/yang), *zhen* (yang), *kun* (yin), *gen* (yang), *kan* (yin/yang) and *sun* (yin).

74 – However, Shao associates the *He Tu* and *Luo Shu* chiefly with the King Wen diagram in *HJJS*, 9.1 ff.

75 – *HJJS*, 7B.1a.

76 – “Dialogue,” sec. 10.

77 – “Dialogue,” sec. 17. “Change” here is Birdwhistell’s translation of *bian*. See her discussion in *Transition*, p. 154.

- 79 – Fung, *A History of Chinese Philosophy* vol. 2: 463–464.
- 80 – *HJJS*, 7A.34b. Wai-lu Hou treats Shao at length as a member of the Chinese idealist tradition, *Weixin shuyi*. See his *Zhongguo sixiang tongshi* (Beijing: Renmin, 1959), pp. 521 ff.
- 81 – Arthur O. Lovejoy, *The Great Chain of Being: A Study of the History of an Idea* (Cambridge: Harvard University Press, 1936), p. 146.
- 82 – Wyatt, Smith, et al., *Sung Dynasty Uses of the I Ching*, p. 123. See Birdwhistell's discussion in *Transition*, pp. 140–144.
- 83 – “Preface to an Edition of Nizolius” (1670), in *L*, p. 129.
- 84 – “Dialogue,” sec. 2.
- 85 – *HJJS*, 8B.28b.
- 86 – *Ibid.*, 8B.27b.
- 87 – “On the True *Theologica Mystica*” (1690), in *L*, p. 368. Shao had similar views: “If it is the spirit of a human being, then it is the spirit of heaven and earth” (*HJJS*, 8B.16b); “Although one is human, one is also heaven” (*ibid.*, 7B.20b).
- 88 – *Ibid.*, 8B.25a.
- 89 – *Ibid.*, 8B.29a.
- 90 – *Ibid.*, 8B.25a.
- 91 – “Dialogue,” sec. 9.
- 92 – “On the Radical Origination of Things” (1697), in *L*, p. 491.
- 93 – “On the Elements of Natural Science” (1682–1684), in *L*, p. 280.
- 94 – *HJJS*, 8B.31b. For a discussion of Shao's view of the place of personal effort in a deterministic cosmos, see my “The Compatibilist Philosophy of Freedom of Shao Yong,” *Journal of Chinese Philosophy* 20 (1993): 279–291.
- 95 – *HJJS*, 7B.19b. In spite of Shao's strictures on divination, he did theorize about looking into the future as a function of reading the Diagram backwards (counterclockwise) (*HJJS*, 8B.2b; Wyatt, Smith, et al., *Sung Dynasty Uses of the I Ching*, p. 109). Birdwhistell shows that Shao had in mind mainly a general understanding of the flow of events rather than predictions of the fates of particular things (Birdwhistell, “The Philosophical Concept of Foreknowledge in the Thought of Shao Yung,” *Philosophy East and West* 39 [1989]: 47–65).
- 96 – “Letter to Remond” (1714), in *L*, p. 657.

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- 97 – Ho Peng Yoke shows that we should not be surprised at this kind of association of science and divination, since the Chinese traditionally viewed their numerical arts (*shu*) (such as *Yijing*) to be predictively powerful for natural processes as well as for human affairs. Thus, novel mechanistic techniques in science seemed to offer nothing new, as far as purpose and results were concerned. See Ho Peng Yoke, "Chinese Science: The Traditional View," *Bulletin of the School of Oriental and African Studies* 54 (1991): 506–519.
- 98 – *HjJS*, 8A.16b.
- 99 – See Joseph Needham, *Science and Civilisation in China*, vol. 3 (Cambridge: Cambridge University Press, 1959), p. 216 ff.
- 100 – Wyatt, Smith, et al. show that Shao speaks of models (*fa*) as inferior to principle (*li*) (*Sung Dynasty Uses of the I Ching*, p. 129).
- 101 – *HjJS*, 8B.29a.
- 102 – *ibid.*, 7B.23b. As Leibniz himself argued (*DNTC*, pp. 113–115), the terms *li*, *taiji*, and *shen* are practically equivalent, even though they have different meanings. *Li* is the rationality of the absolute harmony of things, the *taiji*. The *taiji* continuously moves toward the creation, and yet the harmony, of all things. *Shen* is the force that drives that movement. Ho Peng Yoke points out that "to the common people [*shen*] means 'spirit', but to the Neo-Confucianists it would be the expansion of *qi*, something rather more natural than supernatural" ("Chinese Science," pp. 517–518).
- 103 – *HjJS*, 7B.2b.
- 104 – *ibid.*, 7B.21b.
- 105 – *ibid.*, 7A.17b.
- 106 – *ibid.*, 8B.16b.